

Lab 11

Announcement

- SE Assignment 5 will be released soon.
 - Review Lecture 16 before the 10 minutes online assignment.
- Next lab, **Thu May 21** – Very important survey
 - Every student must arrive on time at **9 am**.
 - Please bring a pen.
 - If you cannot attend, you must have an **official excuse**; otherwise **points will be deducted** from your grade.
- **Presentations** will be held on **May 26 to 28**
 - Take place during the lab and SE class time.
 - Sign up with your teacher.

Tasks

1. Teachers

- A. Explain Test Plan Template
- B. Explain Test Report Template

2. Students (after your code is running)

- A. Prepare test plan
- B. Prepare test report

Teachers Issue 1A: Explain Test Plan

- Test plan template
 - Available on iSpace

Teachers Issue 1B: Explain Test Report Template

- Test Report Template
 - Available on iSpace
 - Report only test cases that contain bugs
 - If no test cases have bugs, write down NONE.

Student Task 2A: Prepare Test Plan

- Use black-box testing on these use cases. Use one table for each use case.
 - Search Student's Info (by Mentor and Coordinator)
 - Edit Interview Record (by Mentor)
 - Search Mentor Info (by Faculty Consultant and Coordinator)
- In Lab 4, you've made transition diagrams for these use cases.
 - See Lecture 14 around slide 8 on how to generate test cases in a transition diagram.

Student Task 2A: Prepare Test Plan

Remark

- In unit testing, we tested a class and the methods in that class.
- In system testing, we test use cases.
 - Each use case may involve several classes.

Student Task 2A:

Prepare Test Plan

Load testing tool: *LoadRunner*

- Do the use case **Edit Interview Record**.
- Choose one of the following versions:
 - Version 12.02 (Installed in computer room for workshop III)
 - » Tutorial available in iSpace
 - Version 2021 (can be downloaded in your own computer)
 - » <https://www.microfocus.com/en-us/products/loadrunner-professional/download>
- Each student reads the document “Reading instructions and questions.pdf”.
- Answer the questions, and submit the answers in iSpace.

Student Task 2A: Prepare Test Plan

- Testing tool: *TestComplete* by SmartBear
- Do the use case: **Search Student's Info**
- Purpose:
 - Students design black-box test cases.
 - As noted in 3 slides back.
 - TestComplete records and repeats the same GUI tests after bug fixes or code changes.

Student Task 2A: Prepare Test Plan

Read / watch these 6 short **tutorials** before doing the use case.

- We listed the doc files, but videos are also available.
- Keyword Tests
<https://support.smartbear.com/testcomplete/docs/keyword-testing/>
- Create / Record a Test
<https://support.smartbear.com/testcomplete/docs/tutorials/getting-started/first-test/web/creating-test.html>
- Analyze the Recorded Test
<https://support.smartbear.com/testcomplete/docs/tutorials/getting-started/first-test/web/analyzing-recorded-test.html>

Student Task 2A: Prepare Test Plan

- Run the Recorded Test

<https://support.smartbear.com/testcomplete/docs/tutorials/getting-started/first-test/web/running-recorded-test.html>

- Analyze Test Results

<https://support.smartbear.com/testcomplete/docs/tutorials/getting-started/first-test/web/analyzing-test-results.html>

- Checkpoints

<https://support.smartbear.com/testcomplete/docs/testing-with/checkpoints/>

Student Task 2A:

Prepare Test Plan

Practice steps:

1. Create a TestComplete project for the web application.
2. Record a Keyword Test for the use case Search Student's Info.
3. Add Checkpoint 1: valid Student ID -> student information is displayed.
4. Add Checkpoint 2: empty/invalid Student ID -> warning message is displayed.
5. Run the test and view the test log.

Submission: take three whole-screen screenshots, with one sentence explanation in each:

- A. The recorded Keyword Test steps,
- B. The added checkpoint steps/settings for the two expected results;
- C. The test log after running the test, showing pass/fail result.

Use an installed copy of TestComplete. See previous 2 slides for instruction pages.

Student Task 2B: Prepare Test Report

- Students prepare the test report for the use cases specified.
- In the **Appendix of Test Report**, please include
 - One Vuser script copy (icon-based view of script or text-based view of script) from the use of *LoadRunner* (similar to pages 17, 18 in the tutorial)

Submission

- **Group:**
 - Each group submit **ONE** copy of **test plan** and **ONE** copy of **test report** to iSpace and GitLab.
 - Follow TestComplete tutorial on slides 9-12. Submit three whole-screen screenshots, each with one sentence explanation in the **appendix** of the **test report**.
 1. recorded Keyword Test steps;
 2. added checkpoint steps/settings;
 3. test log after running the test.
- **Deadline: Sat 11:55 pm, 16 May 2026**

Submission

- **Each Student:**
 - Practice according to the *LoadRunner* tutorial and answer the questions in the document “*Reading instructions and questions.pdf*”.
 - Submit your answers to iSpace only.
- **Deadline: Sat 11:55 pm, 16 May 2026**

Submission

Five grades will be given

- Test Plan
 - Text (Workshop rubric)
 - Test cases (SE rubric)
- Test Report
 - Text (Workshop rubric)
- Testing Tool
 - Loadrunner (Workshop rubric)
 - Test Complete (Workshop rubric)